

Repo Link: <https://github.com/ibrahimmaq14/SE-Labs-PES1UG24CS578/tree/main/LAB4>

SE Lab 4 - Vibe Coding Reconstructed Codex Dialogue

Illustrative reconstruction based on the completed Dig Dug work; this is not an export of the actual chat.

User

I am working on the Dig Dug game in the assigned repository. Read its README and game.py. The first task is to fix the enemy pathfinding bug: once tunnels connect an enemy to the player, it should choose the shortest cleared route instead of wandering along a longer one.

Assistant

Tool: `view_file`

`29_digDug/README.md`

Reviewed the controls, expected behavior, and the four lab tasks.

Tool: `view_file`

`29_digDug/game.py`

Found that `bfs_path` removed nodes from the right side of its deque.

Tool: `code_edit`

`LAB4/Updated Code 29_digDug/game.py`

Changed `queue.pop()` to `queue.popleft()` so the search runs in breadth-first order.

I fixed `bfs_path` to process the oldest queued cell first. Enemies now use the shortest route through cleared cells. In the same tunnel layout, the original route was 10 steps and the corrected route is 8 steps.

User

The second task is to implement `dirt_color(row)`. Give underground rows distinct color bands so the depth is easier to see, while keeping the sky and open tunnels unchanged.

Assistant

Tool: `code_edit`

`LAB4/Updated Code 29_digDug/game.py`

Added four RGB dirt colors for rows 1-3, 4-6, 7-9, and 10-13.

The dirt now changes color in three-row bands as the player digs deeper. Every value is a valid RGB color, and the existing sky and cleared tunnel drawing remain intact.

User

The third task is to implement `on_enemy_popped(enemy, score)`. Add a bonus for popping enemies deeper underground, and make sure the extra points actually appear in the game's score.

Assistant

Tool: `code_edit`

`LAB4/Updated Code 29_digDug/game.py`

Made `on_enemy_popped` return the score with a 50-point bonus for each four rows of depth, then applied that return value in `pump()`.

Tool: `test`

`LAB4/test_game.py`

Verified that a row-8 enemy pops after four pumps and the score reaches 500.

A deep enemy pop now adds a depth bonus to the displayed score. At row 8, the normal 400 points plus the 100-point bonus give 500 points, and the enemy is removed after the fourth pump.

User

The fourth task is to implement `enemy_speed_multiplier(level)`. Make enemies gradually faster on later levels but keep the original timing on level 1.

Assistant

Tool: `code_edit`

`LAB4/Updated Code 29_digDug/game.py`

Implemented a 1x base multiplier plus 0.1x for each level after level 1.

Tool: `test`

`LAB4/test_game.py`

Checked that level 3 uses a 1.2x multiplier and a shorter enemy step delay.

Enemy movement now speeds up with each level. Level 1 remains at 1x, while level 3 runs at 1.2x. The existing ghost slowdown still applies to ghost enemies.

Assigned source: https://github.com/SETAPESU26/29_digDug